tory and neurological health," adds Dr Ingale. The worry with all Long Covid symptoms, however, remains the same—how long will they last and how can it be diagnosed on time?

THE DIAGNOSIS DIFFICULTY

All infections commonly lead to post-recovery weakness. As Dr Chandrashekhar T., director of critical care at Mumbai's Fortis Hiranandani Hospital, says: "There is nothing unique in the post-Covid weakness as such symptoms are common post recovery from many other diseases, like malaria, typhoid, dengue and influenza. The body needs time to recover from all illness and in the initial days after symptoms subside, there will be weakness." What is different with Long Covid fatigue is how long it lasts. In December 2020, the Centers for Disease Control and Prevention (CDC) in the US followed nearly 300 people who were Covid positive for three weeks and found that nearly half still reported chronic fatigue symptoms two months after testing negative. A few months earlier, in September 2020, an Irish study supported by the Wellcome Trust studied 128 patients and found that after 10 weeks, 52 per cent reported persistent fatigue and 31 per cent had not returned to work. The study could find no way to link the length or severity of the fatigue to any laboratory or clinical markers. In other words, there is no investigative test to diagnose how long or how severe a patient's fatigue will be.

Harder still to diagnose are Covid's impact on the brain and mind. An observational study, published in *The Lancet Psychiatry* journal, looked at the electronic medical records of over half a million patients in the US and found that they were 16 per cent more likely to develop a psychological or neurological disorder after Covid than after other respiratory infections, and 44 per cent more likely as compared to people recovering from the flu. Mood-and anxiety-related or psychotic disorders affected 24 per cent of all Covid patients, dementia was diagnosed in 0.7 per cent; and delirium in 5 per cent. "The mental manifestation of Long Covid is easy to misdiagnose because you don't have clear-cut physical symptoms. Memory loss, confusion, anxiety can easily be linked to other factors or not taken seriously by family members," says Dr Kedar Tilwe, a Mumbai-based psychiatrist.

Now a first-of-its-kind study has proved that Covid not only causes neurological issues, but also 'alters' the brain itself. The UK Biobank project followed 500,000 people for about 15 years and has a database of scans recorded before the pandemic. The scientists rescanned 401 participants around four and a half months after their infection and found that the overall brain size in infected participants had shrunk by 0.2-2 per cent. There was a loss in the grey matter in the olfactory areas linked to smell, and regions linked to memory. Out of the respondents, 96 per cent had had mild Covid, 384 participants had been asymptomatic. The study, published in March 2022 in the journal *Nature,* was not able to conclude whether these changes are reversible.

> A study found that Covid patients were 16 per cent more likely to **develop a psychological or neurological disorder than patients of other respiratory infections**

"The correct treatment, if given on time, can reduce the impact of any neurological damage in the brain—which often happens due to the severity of inflammation caused by Covid in the body. Inflammation is dangerous for all organs, not just the brain," says Dr Sheela Chakravarty, an internal medicine specialist from Bengaluru.

In the case of Tripti Gurha (*name changed on request*), the effects of Long Covid made themselves known too late. During her Covid illness, the 39-year-old from Pune had high fever for about four days. "It had been relentless," says her engineer husband Varun Gurha, 41. His wife had high d-dimer (a marker for the amount of clotting in the blood) during her illness and the couple had assumed the parameter would reduce automatically after recovery and did no follow-up blood tests to confirm the same. "She stopped the blood thinners after the prescribed duration. There were no other symptoms, not even weakness or brain fog, so we thought she was fine." A month after recovery in February 2022, Tripti collapsed at her office due to a brain stroke and has been in a coma since.

"Sudden deaths or strokes or heart attacks are being seen in those recovering from Covid," says Delhi-based cardiologist Dr Rajeev Rathi. A study titled, 'Long Term Cardiovascular Outcomes of Covid 19', published in *Nature Medicine* on February 7, 2022, the first large study done to assess the risk of such attacks in the long term, has shown patients to be at risk from over 20 different heart and vessel maladies even a year after recovery. Over 50 per cent of those studied were more likely to have a stroke than those without Covid and the risk of heart failure was 72 per cent. The study looked at the health records of 11 million US veterans and found even those with mild Covid to be at risk from heart attacks, strokes, and heart disease. "As of now, there is no way to predict when a person has recovered from Long Covid or is out of the danger zone. Inflammation within the body can have sudden manifestations even a significant time after the disease itself has subsided. And